Surface tension and the self-similar growth parameter α in immiscible Rayleigh–Taylor mixing

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Abstract

High-resolution simulations of immiscible Rayleigh–Taylor (RT) mixing show a late-time growth parameter α that is larger than in the miscible case and that increases as the initial Weber number We_0 decreases [23, 24, 25]. These notes examine the We_0 dependence of α using the self-similar moment analysis of Ristorcelli & Clark [19], the mass-flux relation of Cook, Cabot & Miller [7], and a modified growth equation that adds a capillary correction to the Ristorcelli–Clark equation. Dimensional analysis alone does not explain the We_0 dependence. Under complete self-similarity, the capillary term in the scalar-flux budget is the inverse instantaneous Weber number, $(\ell_\sigma/h)^2/2\alpha$, so surface tension enters as a correction that decays as the layer grows; this correction is proportional to We_0^{-1} and reduces α. A dependence that increases α requires a closure for the Ristorcelli–Clark constant C_o; a spectral closure gives $\alpha - \alpha_\infty \propto \mathrm{We}_0^{-(3-D_s)/2}$, with D_s the interface fractal dimension. In the simulations, compared at equal mixing width, the excess growth parameter over the case without surface tension follows a single curve, $\alpha - \alpha_\infty \approx 8.9\,\ell_\sigma/h$, which corresponds to a $\mathrm{We}_0^{-1/2}$ dependence at fixed h/λ_0 and to $\nu = 1$ in the modified growth equation.

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1 Introduction

The self-similar growth of a Rayleigh–Taylor mixing layer is usually written $h \approx \alpha \mathrm{A_t} g t^2$ [27, 11]. Ristorcelli & Clark [19] (hereafter RC04) showed that self-similarity of the second-moment equations leads to the ordinary differential equation $\dot{h}^2 = \mathrm{A_t} g C_\mathrm{o} h$ (their eq. 3.6), whose solution contains a virtual origin, and that the constant C_o, and hence $\alpha = C_\mathrm{o}/4$, is a functional of the unclosed self-similar profiles rather than a universal constant. Cook, Cabot & Miller [7] (hereafter CCM04) related the growth rate to the net mass flux through the mid-plane and argued that immiscible fluids, for which the density fluctuations take their largest possible value, should grow faster than miscible ones.

Experiments and simulations support this. Experiments, many of them with immiscible fluids, give growth parameters larger than the simulations of miscible mixing in the Alpha-Group comparison [11, 10]. In experiments at $\mathrm{A_t} \approx 0.5$ that compared the two directly, miscibility reduced α by about a factor of two [20]. Recent simulations of immiscible RT turbulence [29, 28] also find growth parameters above typical miscible values, with α increasing with surface tension [28].

Those simulations solve the incompressible equations with low-order finite-volume, volume-of-fluid methods.

The simulations considered here instead solve the compressible multi-component equations with the high-order compact finite-difference code Miranda [6], using the localized artificial diffusivity method with interface sharpening of Brill, Olson & Bokman [3] and a continuum surface force [2]. At $\mathrm{A_t} = 0.5$, the late-time growth parameter is $\alpha_{\mathrm{imm}} \approx 0.039$ for immiscible fluids without surface tension and $\alpha_{\mathrm{misc}} \approx 0.028$ for miscible fluids [23, 24, 25]. With finite surface tension, the late-time α increases further as the initial Weber number decreases. This trend was reported as approximately $\alpha \propto \mathrm{We}_0^{-1/2}$ on dimensional grounds and fitted as $\alpha \approx 0.49\,\mathrm{We}_0^{-0.35}$ [24]. The macroscopic growth of these simulations is described in Ref. [25], and their mesoscale mixing and spectra in Ref. [26].

Linear theory predicts the opposite tendency: surface tension stabilizes short waves and reduces the exponential growth rate [4, 21]. The question addressed here is what can be said about the We_0 dependence of α from dimensional analysis and from the self-similar analysis of RC04. Section 2 defines the problem, the growth diagnostics and the relevant dimensionless groups. Section 3 extends the RC04 analysis to immiscible fluids with surface tension. Section 4 develops a modified growth equation and relates its exponent to the interface fractal dimension. Section 5 develops the mass-flux formulation of CCM04 for the full width used in the simulations. Section 6 summarizes the resulting scaling in We_0, and Section 7 discusses the simulation results.

2 Problem definition

2.1 Parameters

Two incompressible, immiscible fluids of densities $\rho_h > \rho_\ell$ are separated by an interface with surface tension coefficient σ, under gravity g acting in the $-z$ direction. Define

$$\bar{\rho} = \tfrac{1}{2}(\rho_h + \rho_\ell)\,, \qquad \Delta\rho = \rho_h - \rho_\ell\,, \qquad \mathrm{A_t} = \frac{\Delta\rho}{2\bar{\rho}}\,. \tag{2.1}$$

The initial perturbation occupies the band of mode numbers $[k_{\min}, k_{\max}]$ in a domain of width L. Following Refs. [24, 25], the reference length, time and Weber number are

$$\lambda_0 = \frac{L}{k_{\max}}\,, \qquad \tau_0 = \Big(\frac{\lambda_0}{\mathrm{A_t} g}\Big)^{1/2}\,, \qquad \mathrm{We}_0 = \frac{\bar{\rho}\, g\, \lambda_0^2}{\sigma}\,. \tag{2.2}$$

In the inviscid limit, the linear growth rate of a mode of wavenumber k is [4, 21]

$$\gamma^2 = \mathrm{A_t} g k - \frac{\sigma k^3}{\rho_h + \rho_\ell} = \mathrm{A_t} g k \left(1 - k^2 \ell_\sigma^2\right), \qquad \ell_\sigma \equiv \Big(\frac{\sigma}{\Delta\rho\, g}\Big)^{1/2}\,, \tag{2.3}$$

so ℓ_σ is the linear capillary cutoff length. In terms of the Weber number,

$$\frac{\ell_\sigma}{\lambda_0} = (2\mathrm{A_t}\mathrm{We}_0)^{-1/2}\,, \tag{2.4}$$

which equals $\mathrm{We}_0^{-1/2}$ at $\mathrm{A_t} = 0.5$.

2.2 Mixing width and growth diagnostics

Let V be the volume fraction of heavy fluid and $\bar{V}(z,t)$ its average over horizontal planes. The mixing width, which equals the total width of a linear profile, is

$$h = 6 \int_{-\infty}^{\infty} \bar{V}(1 - \bar{V})\, \mathrm{d}z\,. \tag{2.5}$$

For immiscible fluids, the mass fraction Y and the volume fraction V coincide pointwise in the pure phases, so (2.5) is the width used in Refs. [24, 25]. Two diagnostics of α are in use:

$$\alpha(t) \equiv \frac{\dot{h}^2}{4\mathrm{A_t} g h}\,, \qquad \alpha_{\mathrm{eff}}(t) \equiv \frac{\mathrm{d}h}{\mathrm{d}(\mathrm{A_t} g t^2)} = \frac{\dot{h}}{2\mathrm{A_t} g t}\,. \tag{2.6}$$

The first is used in Refs. [24, 25]; the second is eq. 3.8 of RC04. For the RC04 solution, written for the full width,

$$h = \alpha \mathrm{A_t} g t^2 + 2t\,(\alpha \mathrm{A_t} g h_0)^{1/2} + h_0\,, \tag{2.7}$$

one finds $\alpha(t) = \alpha$ for all t, whereas $\alpha_{\mathrm{eff}} = \alpha\,[1 + (h_0/\alpha \mathrm{A_t} g t^2)^{1/2}]$ retains the virtual origin. Integrating the definition of $\alpha(t)$ gives

$$\sqrt{h(t)} = \sqrt{h(0)} + \int_0^t \big(\alpha(t')\,\mathrm{A_t} g\big)^{1/2}\,\mathrm{d}t'\,. \tag{2.8}$$

2.3 Dimensional considerations

With ρ_h, ρ_ℓ, g, σ, the viscosity, λ_0, the shape of the initial spectrum and the domain size as parameters, Buckingham's theorem gives

$$\frac{h}{\lambda_0} = \Phi\Big(\frac{t}{\tau_0};\, \mathrm{A_t},\, \mathrm{We_0},\, \mathrm{Re_0},\, \ldots\Big)\,, \qquad \alpha(t) = \Psi\Big(\frac{t}{\tau_0};\, \mathrm{A_t},\, \mathrm{We_0},\, \mathrm{Re_0},\, \ldots\Big)\,, \tag{2.9}$$

where the dots denote the initial-condition and domain parameters. Dimensional analysis therefore admits any dependence on $\mathrm{We_0}$; it does not select an exponent. Three relations are nevertheless useful.

First, $\sigma/\bar{\rho}$ has dimensions $\mathrm{L^3T^{-2}}$. Combined with g, it gives the length ℓ_σ, and combined with λ_0, it gives the capillary time $t_\sigma = (\bar{\rho}\lambda_0^3/\sigma)^{1/2}$, so that

$$\mathrm{We_0} = \frac{1}{\mathrm{A_t}}\Big(\frac{t_\sigma}{\tau_0}\Big)^2\,. \tag{2.10}$$

The statement that $\mathrm{We_0} \propto 1/\sigma$ "scales as t^2" expresses that $\mathrm{We_0}$ is the square of a ratio of time scales; it does not imply a power law for α.

Second, the Weber number based on the instantaneous layer width and growth velocity is

$$\mathrm{We}(t) \equiv \frac{\bar{\rho}\,\dot{h}^2 h}{\sigma} = 4\mathrm{A_t}\,\alpha\,\mathrm{We_0}\Big(\frac{h}{\lambda_0}\Big)^2 = 4\mathrm{A_t}\,\alpha^3\,\mathrm{We_0}\Big(\frac{t}{\tau_0}\Big)^4\,, \tag{2.11}$$

where the last form uses $h/\lambda_0 = \alpha(t/\tau_0)^2$ in the self-similar regime. With (2.4),

$$\Big(\frac{\ell_\sigma}{h}\Big)^2 = \frac{2\alpha}{\mathrm{We}(t)}\,. \tag{2.12}$$

At fixed t/τ_0, $\mathrm{We}(t) \propto \alpha^3 \mathrm{We_0}$, so a power law in $\mathrm{We}(t)$ appears as a power law in $\mathrm{We_0}$ when results are compared at fixed time, with the same exponent only if α does not depend on $\mathrm{We_0}$.

Third, if surface tension modifies the late-time growth through a velocity scale set by capillarity and buoyancy alone, independently of λ_0, that scale is $U_\sigma \propto (\mathrm{A_t} g \ell_\sigma)^{1/2}$, the only velocity formed from $\mathrm{A_t} g$ and $\sigma/\Delta\rho$. Adding U_σ^2 to the self-similar $\dot{h}^2 = 4\alpha_\infty \mathrm{A_t} g h$ gives

$$\dot{h}^2 = 4\mathrm{A_t} g\,(\alpha_\infty h + K\ell_\sigma) \quad \twoheadrightarrow \quad \alpha(t) = \alpha_\infty + K\,\frac{\ell_\sigma}{h} = \alpha_\infty + K\Big(\frac{2\alpha}{\mathrm{We}(t)}\Big)^{1/2}\,, \tag{2.13}$$

with K a constant. At fixed h/λ_0, the correction is proportional to $(2\mathrm{A_t We_0})^{-1/2}$. This is the form in which a $\mathrm{We}_0^{-1/2}$ dependence is consistent with dimensional analysis: as a correction that decays as h^{-1}, not as $\alpha \propto \mathrm{We}_0^{-1/2}$. The latter would give $\alpha \nrightarrow 0$ as $\sigma \nrightarrow 0$, contrary to $\alpha_{\mathrm{imm}} \approx 0.039$ at $\mathrm{We_0} = \infty$.

3 Self-similar analysis with immiscibility and surface tension

3.1 Governing equations

Following RC04, the Boussinesq form is used for the analysis; Section 5 treats variable density through the mass flux. Let $c = +1$ in the heavy fluid and $c = -1$ in the light fluid, so that $\rho = \bar{\rho}(1 + \mathrm{A_t} c)$. With the pressure divided by $\bar{\rho}$,

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} = -\nabla p - \mathrm{A_t} g\, c\, \mathbf{e}_z + \frac{\mu}{\bar{\rho}} \nabla^2 \mathbf{u} + \nabla \cdot \mathbf{T}^\sigma \,, \tag{3.1}$$

$$\frac{\partial c}{\partial t} + \mathbf{u} \cdot \nabla c = 0 \,, \qquad \nabla \cdot \mathbf{u} = 0 \,, \tag{3.2}$$

where $\mathbf{u} = (u, v, w)$ is the velocity, w is its component along the upward unit vector $\mathbf{e}_z$, μ is the dynamic viscosity, and the capillary stress tensor is [14]

$$\mathbf{T}^\sigma = \frac{\sigma}{\bar{\rho}} \left(\mathbf{I} - \hat{\mathbf{n}}\hat{\mathbf{n}} \right) \delta_s \,, \qquad \hat{\mathbf{n}} = \frac{\nabla c}{|\nabla c|} \,, \tag{3.3}$$

and δ_s is the surface delta function. Since $\nabla \cdot [(\mathbf{I} - \hat{\mathbf{n}}\hat{\mathbf{n}})\delta_s] = \kappa\, \hat{\mathbf{n}}\, \delta_s$, with $\kappa = -\nabla \cdot \hat{\mathbf{n}}$ the curvature of the interface, the force $\mathbf{f} \equiv \nabla \cdot \mathbf{T}^\sigma = (\sigma/\bar{\rho})\, \kappa\, \hat{\mathbf{n}}\, \delta_s$ is the continuum surface force of Brackbill et al. [2]. Angle brackets denote averages over horizontal planes, and $c = C + c'$ with $C = \langle c \rangle$. Incompressibility and the walls give $\langle w \rangle = 0$.

3.2 The immiscibility constraint

Because $c^2 = 1$ everywhere,

$$\langle c'c' \rangle = 1 - C^2 \,. \tag{3.4}$$

RC04 define the molecular mixing parameter $\theta_m = 1 - \langle c'c' \rangle / [(1 - C)(1 + C)]$ (their eq. 4.12, in which their c is the fluctuation c'); by (3.4), it vanishes for immiscible fluids. Their scalar-variance equation (eq. 2.9) is then satisfied automatically, since the scalar dissipation is zero and $\langle wc'c' \rangle = -2C \langle wc' \rangle$. The relation $\langle c'c' \rangle_{\mathrm{o}} = C_{\mathrm{o}}^2$ that RC04 obtain from the similarity groups (their eq. 3.4) is replaced, on the mid-plane where $C = 0$, by

$$\langle c'c' \rangle_0 = 1 \,. \tag{3.5}$$

Here the subscript 'o' follows RC04: it marks the temporal amplitude $F_{\mathrm{o}}(t)$ in the self-similar form $F(z,t) = F_{\mathrm{o}}(t)\, f(\eta)$, which is proportional to the mid-plane value for an even profile and is set by integrals of the profile for an odd one. The amplitude C_{o} of the mean indicator is constant in the turbulent growth phase (RC04 §3) and is the growth constant in (3.14). The subscript '0' on a statistic denotes its value on the mid-plane $z = 0$; on λ_0, τ_0, $\mathrm{We_0}$, $\mathrm{Re_0}$ and h_0, it denotes initial or reference values. For a miscible flow, $\langle c'c' \rangle_0 = 1 - \theta_0$, with θ_0 the mid-plane mixedness. This is the only place where miscibility enters the self-similar analysis.

3.3 Mean scalar equation

Averaging (3.2) gives

$$\frac{\partial C}{\partial t} + \frac{\partial}{\partial z}\langle w c' \rangle = 0 \tag{3.6}$$

(RC04 eq. 2.7 without molecular diffusion). With the self-similar form $C = f_C(\eta)$, $\eta = z/h$ (RC04 eq. 3.1), integration from $-\infty$ to the mid-plane gives

$$\langle w c' \rangle_0 = -\frac{\mathrm{d}}{\mathrm{d}t}\int_{-\infty}^{0} (1 + C)\,\mathrm{d}z = -\frac{2}{R_f}\,\dot{h}\,, \qquad \frac{2}{R_f} = \int_{-\infty}^{0} \big(1 + f_C(\eta)\big)\,\mathrm{d}\eta\,. \tag{3.7}$$

This corresponds to the RC04 similarity group $\langle u_3 c\rangle_\mathrm{o}/(\dot{h} C_\mathrm{o})$. Here R_f is the mass-flux shape factor (5.3); the integral equals $2/R_f$ because $1 + C = 2\bar{V}$ and $\rho' = (\Delta\rho/2)\,c'$.

3.4 Mid-plane scalar-flux budget

Multiplying the vertical component of (3.1) by c', the indicator equation by w, adding and averaging gives (as in RC04 eq. 2.8; see also Ref. [17])

$$\frac{\partial}{\partial t}\langle w c' \rangle + \frac{\partial}{\partial z}\langle w w c' \rangle = -\langle w w \rangle \frac{\partial C}{\partial z} - \mathrm{A_t} g \langle c' c' \rangle - \Big\langle c' \frac{\partial p}{\partial z} \Big\rangle + \langle c' f_z \rangle + \frac{\mu}{\bar{\rho}} \langle c' \nabla^2 w \rangle\,. \tag{3.8}$$

The fluctuating pressure satisfies

$$\nabla^2 p' = -\mathrm{A_t} g\, \frac{\partial c'}{\partial z} - \frac{\partial^2 (u_i u_j - \langle u_i u_j \rangle)}{\partial x_i\, \partial x_j} + \nabla \cdot (\mathbf{f} - \langle \mathbf{f} \rangle)\,, \tag{3.9}$$

and is split into buoyancy, nonlinear and capillary parts, $p' = p_B + p_N + p_\sigma$. The buoyancy part cancels a fraction χ of the buoyancy production:

$$-\mathrm{A_t} g \langle c' c' \rangle - \Big\langle c' \frac{\partial\, p_B}{\partial z} \Big\rangle = -\mathrm{A_t} g\,(1 - \chi) \langle c' c' \rangle\,, \qquad \chi \equiv \frac{1}{\langle c' c' \rangle} \Big\langle c'\, \frac{\partial^2}{\partial z^2} \nabla^{-2} c' \Big\rangle\,. \tag{3.10}$$

For statistically homogeneous fields, $\chi = \sum_{\mathbf{k}} (k_z^2/k^2) |\hat{c}_{\mathbf{k}}|^2 / \sum_{\mathbf{k}} |\hat{c}_{\mathbf{k}}|^2$, so $\chi \to 0$ for vertical columns, $\chi = 1/3$ for isotropic structures and $\chi \to 1$ for horizontal layers. The capillary force enters only through its solenoidal part, $\mathcal{P}\mathbf{f} = \mathbf{f} - \nabla p_\sigma$; the Laplace pressure of a drop in equilibrium is a gradient and does not contribute.

Let $F = -\langle w c' \rangle_0 = 2\dot{h}/R_f$ from (3.7). Each mid-plane term in (3.8) is normalized by $\dot{h}^2/h$, with signs chosen so that a positive value is a sink of F:

$$\begin{gathered} d_T = -\frac{h}{\dot{h}^2}\,\frac{\partial}{\partial z}\langle w w c' \rangle\Big|_0\,, \qquad d_N = -\frac{h}{\dot{h}^2}\Big\langle c' \frac{\partial\, p_N}{\partial z} \Big\rangle_0\,, \qquad d_\mu = \frac{h}{\dot{h}^2}\,\frac{\mu}{\bar{\rho}} \langle c' \nabla^2 w \rangle_0\,, \\ d_\sigma = \frac{h}{\dot{h}^2} \langle c' (\mathcal{P}\mathbf{f})_z \rangle_0\,, \qquad \hat{w} = \frac{\langle w w \rangle_0}{\dot{h}^2}\,, \qquad f_C'(0) = h \frac{\partial C}{\partial z}\Big|_0\,. \end{gathered} \tag{3.11}$$

Evaluating (3.8) on the mid-plane gives the ordinary differential equation

$$\frac{2}{R_f}\,\ddot{h} = \mathrm{A_t} g\,(1 - \chi_0) \langle c' c' \rangle_0 - \Big(\hat{D} - \frac{1}{R_f} + d_\sigma\Big)\frac{\dot{h}^2}{h}\,, \qquad \hat{D} = \frac{1}{R_f} + d_T + d_N + d_\mu - \hat{w}\, f_C'(0)\,. \tag{3.12}$$

When $\dot{h}^2 \propto h$, $\ddot{h} = \dot{h}^2/2h$, and (3.12) reduces to

$$\dot{h}^2 = \mathrm{A_t} g\, h\, \frac{\langle c' c' \rangle_0\,(1 - \chi_0)}{\hat{D} + d_\sigma}\,. \tag{3.13}$$

In the absence of the capillary term, this is RC04 eq. 3.6, $\dot{h}^2 = \mathrm{A_t} g C_\mathrm{o} h$, with

$$C_\mathrm{o} = \frac{\langle c'c' \rangle_0 \, (1 - \chi_0)}{\hat{D}} \,, \qquad \alpha_\infty = \frac{C_\mathrm{o}}{4} \,. \tag{3.14}$$

Equation (3.14) writes out the constant that RC04 describe as a functional of the unclosed profiles: a purity factor $\langle c'c' \rangle_0$, equal to $1 - \theta_0$ for miscible fluids and to unity for immiscible fluids by (3.5); a buoyancy-efficiency factor $1 - \chi_0$; and a normalized inertia and drag $\hat{D}$. Each term is a mid-plane statistic of the $\langle wc' \rangle$ budget. RC04 use the half-width; for the full width (2.5), the same relation $\alpha = C_\mathrm{o}/4$ holds with C_o defined through $\dot{h}^2 = \mathrm{A_t} g C_\mathrm{o} h$.

3.5 Similarity group of the capillary term

In RC04, every energy-containing quantity scales with h. Applying this to the interface, the curvature κ and the surface density Σ that contribute to $\langle c'(\mathcal{P}\mathbf{f})_z \rangle$ both scale as $1/h$, so $\langle c'(\mathcal{P}\mathbf{f})_z \rangle_0 \sim (\sigma/\bar{\rho})/h^2$. By (3.11), (2.11) and (2.12),

$$d_\sigma \sim \frac{\sigma}{\bar{\rho} \, h \, \dot{h}^2} = \frac{1}{\mathrm{We}(t)} = \frac{1}{2\alpha} \Big(\frac{\ell_\sigma}{h} \Big)^2 \,. \tag{3.15}$$

The capillary group is the inverse instantaneous Weber number; it is also the capillary term of (2.3) evaluated at $k \sim 1/h$. At the end of the simulations, $\mathrm{We}(t) \approx (3\text{–}7) \times 10^3$ (Figure 8b), so $d_\sigma \sim 10^{-4}$ while $\hat{D} = \mathcal{O}(1)$.

Small-scale interfacial structure does not change this estimate at leading order. Turbulence cannot deform the interface at scales below the Hinze scale [13, 12]

$$\ell_H = C_H \Big(\frac{\sigma}{\bar{\rho}} \Big)^{3/5} \varepsilon^{-2/5} \,, \qquad C_H \approx 0.725 \,, \tag{3.16}$$

and the capillary force on structures near ℓ_H is predominantly balanced by Laplace pressure. With the inviscid-cascade scaling of RC04 (their eq. 3.14), $\varepsilon_0 = C_\varepsilon \dot{h}^3/h$, and $\dot{h}^2 = 4\alpha \mathrm{A_t} g h$,

$$\frac{\ell_H}{h} = K_H \Big(\frac{\ell_\sigma}{h} \Big)^{6/5} \,, \qquad \frac{\ell_H}{\ell_\sigma} = K_H \Big(\frac{\ell_\sigma}{h} \Big)^{1/5} \,, \qquad K_H = C_H \, C_\varepsilon^{-2/5} (2\alpha)^{-3/5} \,. \tag{3.17}$$

For $\alpha = 0.04$ and $C_\varepsilon = 1$, $K_H \approx 3.3$. The small-scale interfacial structure therefore also separates from h as the layer grows; see also Ref. [5].

The Hinze scale evaluated with the dissipation rate of the simulations [26] follows this scaling (Figure 1). For $6 \le t/\tau_0 \le 9.4$, it decreases as $h^{-0.11}$ to $h^{-0.35}$ for the three Weber numbers, around the $h^{-1/5}$ of (3.17), and a fit with that slope gives $K_H \approx 2.3$; with the late-time values of α in the simulations, 0.08–0.13 rather than 0.04, (3.17) gives $K_H \approx 1.6\text{–}2.2$. At $t = 9.4\tau_0$, $\ell_H/\ell_\sigma \approx 0.8\text{–}0.9$ and $\ell_H \approx 0.6\text{–}1.1 \, \Delta x$ (Section 7.6), so the Hinze scale, the capillary length and the grid spacing are comparable. Across Weber numbers, ℓ_H varies as $\mathrm{We}_0^{-0.45}$ rather than $\mathrm{We}_0^{-3/5}$, so the dissipation rate also depends on We_0.

3.6 Consequences of self-similarity

The quantities $\sigma/\bar{\rho}$ and $\mathrm{A_t} g$ define the length ℓ_σ but no dimensionless group. A second length is needed, and under complete self-similarity the only one available is h, which grows with time; the initial wavelength λ_0 is forgotten. If a self-similar state exists with h as its only outer length, then $d_\sigma \to 0$ and $\alpha \to \alpha_\infty$ given by (3.14), with no explicit dependence on σ. Complete self-similarity therefore excludes a persistent power law $\alpha \propto \mathrm{We}_0^{-m}$.

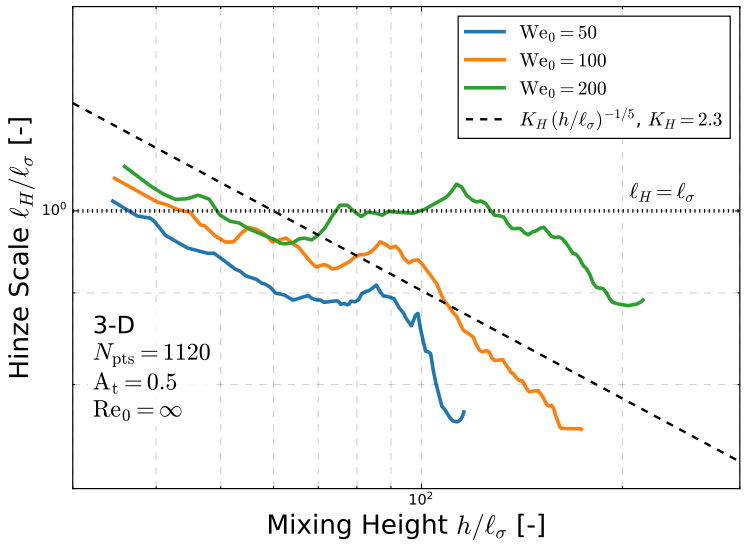


Figure 1: Hinze scale, evaluated with the dissipation rate of the simulations [24, 26], relative to the capillary length against h/ℓ_σ after the transition, with the scaling $K_H(\ell_\sigma/h)^{1/5}$ of (3.17) fitted with $K_H = 2.3$ (dashed). The dotted line marks $\ell_H = \ell_\sigma$; below it the inner cutoff of Section 6.2 is ℓ_H.

Two further lines of analysis are possible. First, the capillary group decays but has not yet vanished; this is incomplete similarity in the sense of Barenblatt [1], and Section 3.7 derives the resulting correction. Second, the profile constants may be inherited from the transition. RC04 note that self-similarity holds $\langle c'c' \rangle_{\mathrm{o}}$ at its value at the start of the turbulent growth phase (their §4), and the same applies to every profile constant in (3.14). During the transition, $h \sim \lambda_0$, and the capillary ratio $\ell_\sigma/\lambda_0 = (2\mathrm{A_t We_0})^{-1/2}$ from (2.4) is not small, so χ_0 and $\hat{D}$ may depend on $\mathrm{We_0}$. The form of that dependence requires a closure, as RC04 state for C_{o}.

3.7 First-order capillary correction

Starting from (3.13), the capillary group is not constant: by (3.15) it decays as a power of h,

$$d_\sigma = \hat{d}_\sigma \Big(\frac{\ell_\sigma}{h}\Big)^{\nu}, \qquad \hat{d}_\sigma \sim \frac{1}{2\alpha}, \qquad \nu = 2\,, \tag{3.18}$$

whereas $\langle c'c' \rangle_0$, χ_0 and $\hat{D}$ take their self-similar values. For $h \gg \ell_\sigma$, $d_\sigma \ll \hat{D}$ and the denominator of (3.13) may be expanded, $(\hat{D} + d_\sigma)^{-1} = \hat{D}^{-1}[1 - d_\sigma/\hat{D} + \mathcal{O}(d_\sigma^2)]$. With (3.14), this gives

$$\dot{h}^2 = \mathrm{A_t} g\, C_{\mathrm{o}}\, h \left[1 - a_\sigma \Big(\frac{\ell_\sigma}{h}\Big)^{\nu}\right], \qquad a_\sigma = \frac{\hat{d}_\sigma}{\hat{D}} > 0\,, \tag{3.19}$$

which is the modified growth equation of Section 4 with $\nu = 2$. The sign of a_σ follows from the restoring character of the capillary force: on the tip of a falling spike, where $c' > 0$, the force points upward; on the cap of a rising bubble, where $c' < 0$, it points downward; in both cases, $c' f_z > 0$, so $d_\sigma > 0$.

Integrating (3.19) for $h \gg \ell_\sigma$ (Appendix A) gives, for the two diagnostics (2.6),

$$\alpha(t) = \alpha_\infty \Big[1 - a_\sigma \Big(\frac{\ell_\sigma}{h}\Big)^\nu\Big] \,, \tag{3.20}$$

$$\alpha_{\rm eff}(t) = \alpha_\infty \Big[1 - a_\sigma \frac{1-\nu}{1-2\nu}\Big(\frac{\ell_\sigma}{h}\Big)^\nu - \frac{c_v}{\sqrt{h}} + \dots\Big] \,, \tag{3.21}$$

where c_v is set by the virtual origin. In $\alpha_{\rm eff}$, the virtual-origin term decays as $h^{-1/2}$ and dominates any correction with $\nu > 1/2$; (3.20) does not contain it. With (2.4) and (2.12), (3.20) reads

$$\alpha(t) = \alpha_\infty \Big[1 - a_\sigma \,(2{\rm A_t We}_0)^{-\nu/2}\Big(\frac{\lambda_0}{h}\Big)^\nu\Big] = \alpha_\infty \Big[1 - a_\sigma \Big(\frac{2\alpha}{{\rm We}(t)}\Big)^{\nu/2}\Big] \,. \tag{3.22}$$

For $a_\sigma > 0$, both diagnostics decrease as σ increases.

3.8 Relaxation toward the self-similar state

Equation (3.12) also describes how a departure from $\dot h^2 \propto h$ decays when the profile constants are fixed. Writing $y = \dot h^2$, so that $\ddot h = \frac{1}{2}\,{\rm d}y/{\rm d}h$, and setting $d_\sigma = 0$, (3.12) becomes the linear equation

$$\frac{{\rm d}y}{{\rm d}h} + \beta\,\frac{y}{h} = R_f {\rm A_t} g (1-\chi_0)\langle c' c'\rangle_0 \,, \qquad \beta = R_f \hat D - 1 \,, \tag{3.23}$$

with solution

$$\dot h^2 = {\rm A_t} g C_{\rm o} h + E\, h^{-\beta} \quad \twoheadrightarrow \quad \alpha(t) - \alpha_\infty = \frac{E}{4{\rm A_t} g}\, h^{-R_f \hat D} \,, \tag{3.24}$$

where E is a constant of integration. With $\alpha_\infty \approx 0.035$, $\chi_0 \approx 1/3$ and $R_f \approx 8.0$, the values without surface tension, (3.14) gives $\hat D \approx 4.8$, so $R_f \hat D \approx 40$. Within the RC04 balance with fixed constants, a departure from self-similar growth decays as a high power of h. A correction that decays only as h^{-1}, as in (2.13), therefore cannot be a transient of (3.12) with fixed constants. It requires either a slow evolution of the constants χ_0 and $\hat D$, or an additional term in the balance that scales as ℓ_σ/h.

4 A modified growth equation

4.1 Statement

The effect of surface tension on the late-time growth is written as a modification of the RC04 equation,

$$\dot h^2 = {\rm A_t} g C_{\rm o} h \Big[1 - a\Big(\frac{\ell_\sigma}{h}\Big)^\nu\Big] \,, \qquad C_{\rm o} = 4\alpha_\infty \,, \tag{4.1}$$

where a is a constant and ℓ_σ is the linear capillary cutoff (2.3). Equation (4.1) has the form of (3.19) with the exponent ν left open. The exponent is related to the fractal dimension D_s of the interface (Section 4.2) through

$$\nu = \frac{6q-2}{5} = \frac{3p-5}{5} = \frac{7-3D_s}{5} \,, \tag{4.2}$$

which gives $\nu = 1/5$ for a smooth interface. For $\nu = 1/5$, the effective growth parameter follows as in RC04,

$$\alpha_{\rm eff}(h) = \frac{C_{\rm o}}{4}\Big[1 - \frac{4a}{3}\Big(\frac{\ell_\sigma}{h}\Big)^{1/5}\Big] \,. \tag{4.3}$$

4.2 The exponent relations

The three forms of (4.2) follow from the properties of a two-phase indicator field. For such a field, the second-order structure function $\mathcal{S}_2(r) = \langle [c(\mathbf{x}+\mathbf{r}) - c(\mathbf{x})]^2 \rangle$ is proportional to the probability that a segment of length r crosses the interface. That probability scales with the surface density measured at resolution r, $\Sigma(r) \propto h^{-1}(h/r)^{D_s-2}$, so $\mathcal{S}_2(r) \propto r^{3-D_s}$ [22]. Defining q by $\mathcal{S}_2 \propto r^{2q}$ and p by the spectrum $E_c(k) \propto k^{-p}$, with $p = 2q+1$,

$$D_s = 3 - 2q = 4 - p \,. \tag{4.4}$$

These relations make the three expressions in (4.2) equal. A smooth interface, $D_s = 2$, has $q = 1/2$ and $p = 2$ (the spectrum of Porod [18]), and gives $\nu = 1/5$. An indicator with the Obukhov–Corrsin spectrum $p = 5/3$ [16, 9] has $D_s = 7/3$.

The value $(6q-2)/5$ is obtained by estimating the capillary term at the Hinze scale: the ratio of capillary to buoyancy pressure at ℓ_H, weighted by the relative amplitude of the indicator fluctuation at that scale,

$$\frac{\text{capillary}}{\text{buoyancy}} \sim \frac{\sigma/\ell_H}{\Delta\rho\, g\, \ell_H} \Big(\frac{\ell_H}{h}\Big)^q = \Big(\frac{\ell_\sigma}{\ell_H}\Big)^2 \Big(\frac{\ell_H}{h}\Big)^q \,. \tag{4.5}$$

Substituting (3.17) and omitting the factors of K_H,

$$\Big(\frac{\ell_\sigma}{h}\Big)^{-2/5} \Big(\frac{\ell_\sigma}{h}\Big)^{6q/5} = \Big(\frac{\ell_\sigma}{h}\Big)^{(6q-2)/5} \,, \tag{4.6}$$

which is (4.2).

4.3 The effective growth parameter

Equation (4.3) is (3.21) with $\nu = 1/5$, for which $(1-\nu)/(1-2\nu) = 4/3$, with the virtual-origin term omitted. The factor $4/3$ appears only in α_{eff}, which divides $\dot{h}$ by the elapsed time and so retains the larger correction from earlier times; for $\alpha(t)$, the measure used in the simulations, the coefficient is a (3.20).

4.4 Interpretation

Equation (4.1) offers a single description of the Weber-number dependence of α across all cases. Decreasing We_0 increases $\ell_\sigma \sim \sigma^{1/2}$ through (2.4). In the simulations, this increases the growth rate and delays self-similarity, and the curves eventually converge (Section 7). The leading-order term of (4.1) is the RC04 equation and gives the standard α. The additional term shifts α relative to the $\mathrm{We}_0 = \infty$ case, and the shift decreases as h grows, so all cases approach α_∞ as $h/\ell_\sigma \to \infty$, consistent with Section 3.6. A shift that decreases with h is what is required for the simulations at different Weber numbers to collapse on a single curve. Compared at equal mixing width, they do correspond to a single function of ℓ_σ/h (Section 7.4).

Three points constrain the form of the correction. First, the sign: with $a > 0$, $\alpha(t)$ (3.20) decreases as ℓ_σ increases, and so does α_{eff} (3.21) for $\nu < 1/2$ or $\nu > 1$, where $(1-\nu)/(1-2\nu) > 0$; the observed increase of α therefore requires $a < 0$. A restoring capillary force gives $a > 0$ (Section 3.7), so a negative a must come from an indirect effect of surface tension on the flow structure. Second, the exponent: the fit in Section 7.4 has $\nu \approx 1$, whereas (4.2) gives $\nu \le 1/5$ for $D_s \ge 2$. Moreover, $\nu > 0$ in (4.2) requires $D_s < 7/3$; at $D_s = 7/3$, the term renormalizes C_o, and for the values $D_s \approx 2.6$–2.7 reported for scalar iso-surfaces in turbulence [22], $\nu < 0$ and the correction would grow with h. Third, the magnitude: with $\nu = 1$ the fit gives $\alpha - \alpha_\infty \approx 8.9\,\ell_\sigma/h$, which in the normalization of (4.1) is $a \approx -8.9/\alpha_\infty \approx -250$. The correction is of order unity relative to α_∞ over the simulated range, and $|a|$ is much larger than unity.

5 Mass-flux formulation

5.1 Growth and mid-plane mass flux

For incompressible phases $\langle\rho\rangle = \rho_\ell + \Delta\rho\bar{V}$, and mean continuity gives

$$\Delta\rho\,\frac{\partial\bar{V}}{\partial t} = -\frac{\partial}{\partial z}\langle\rho w\rangle\ . \tag{5.1}$$

Differentiating (2.5) and integrating by parts, with $\langle\rho w\rangle \to 0$ as $|z| \to \infty$,

$$\dot{h} = -\frac{12}{\Delta\rho}\int_{-\infty}^{\infty}\langle\rho w\rangle\,\frac{\partial\bar{V}}{\partial z}\,\mathrm{d}z\ . \tag{5.2}$$

For immiscible incompressible phases, $\langle w\rangle = 0$, so $\langle\rho w\rangle = \langle\rho' w'\rangle$. CCM04 obtain the corresponding result for their entrainment length h_C (their eqs. 3.5 and 7.3), and approximate it as $\dot{h}_C \approx -4\langle\rho w\rangle_0/\Delta\rho$ (their eq. 7.4).

Define the shape factor

$$R_f \equiv \frac{\Delta\rho\,\dot{h}}{|\langle\rho' w'\rangle_0|}\ , \tag{5.3}$$

twice the "scaling ratio" reported in Refs. [24, 25], which uses the half-width $h/2$. For a self-similar profile, $\langle\rho w\rangle_0 = -\Delta\rho\,(\mathrm{d}/\mathrm{d}t)\int_{-\infty}^{0}\bar{V}\,\mathrm{d}z$, so R_f depends only on the shape of $\bar{V}$:

$$R_f = \frac{h}{\int_{-\infty}^{0}\bar{V}\,\mathrm{d}z} = \frac{4h}{h_C} = \begin{cases} 8 & \text{linear profile}\ , \\ 6\sqrt{2} \approx 8.49 & \text{error-function profile}\ , \\ 6/\ln 2 \approx 8.66 & \tanh\ \text{profile}\ . \end{cases} \tag{5.4}$$

The value $R_f = 4$ corresponds to the entrainment length of CCM04; for the full width (2.5), the self-similar value is 8.0–8.7. Without surface tension, R_f rises into the self-similar range at late times, with a mean of 8.0 over $8 \le t/\tau_0 \le 9.4$, the value for a linear profile; with surface tension it settles at 9.7–10.7, above that range, and in the miscible case it is still increasing (Figure 2, Table 1).

5.2 Decomposition of the growth parameter

Following CCM04 (their eq. 7.5), write $\langle\rho' w'\rangle_0 = -C_r\rho_{\mathrm{rms}}w_{\mathrm{rms}}$, with $\rho_{\mathrm{rms}} = \Delta\rho\,[\bar{V}_0(1-\bar{V}_0)(1-\theta_0)]^{1/2}$, where θ_0 is the mid-plane mixedness, and define $\mathrm{Fr}_0^2 = w_{\mathrm{rms}}^2/(\mathrm{A_t}gh)$. With $\bar{V}_0 = 1/2$, (2.6) and (5.3) give

$$\alpha = \frac{R_f^2}{16}\,C_r^2\,(1-\theta_0)\,\mathrm{Fr}_0^2\ . \tag{5.5}$$

The effective Atwood number of CCM04 (their eq. 3.14) is $\mathrm{A_{eff}} = \mathrm{A_t}(1-\theta_0)^{1/2}$ in this notation. Their closure $w_{\mathrm{rms}} = C_w(\mathrm{A_{eff}}gh)^{1/2}$ (eq. 7.8) gives $\mathrm{Fr}_0^2 = C_w^2(1-\theta_0)^{1/2}$ and hence

$$\alpha = \frac{R_f^2}{16}\,C_r^2C_w^2\,(1-\theta_0)^{3/2} = \frac{R_f^2}{16}\,C_r^2C_w^2\Big(\frac{\mathrm{A_{eff}}}{\mathrm{A_t}}\Big)^3\ , \tag{5.6}$$

which reduces to CCM04 eq. 7.12 for $R_f = 4$.

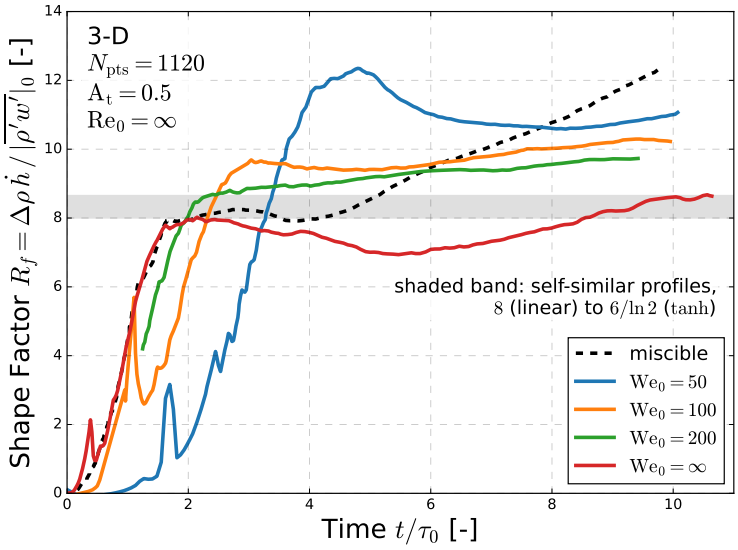


Figure 2: Mass-flux shape factor $R_f = \Delta\rho\,\dot{h}/|\langle\rho' w'\rangle_0|$ for the full width (2.5), twice the scaling ratio of Refs. [24, 25]. The gray band marks the self-similar values (5.4).

Table 1: Decomposition (5.5) of the growth parameter, averaged over $8 \le t/\tau_0 \le 9.4$, with R_f for the full width (2.5) and θ_0 from the mid-plane density fluctuation (3-D, $N_{\rm pts} = 1120$, $\mathrm{A_t} = 0.5$, $\mathrm{Re}_0 = \infty$).

case	α	R_f	θ_0	$C_r^2\mathrm{Fr}_0^2 = 16\alpha/[R_f^2(1-\theta_0)]$	$C_r\mathrm{Fr}_0$
$\mathrm{We}_0 = 50$	0.132	10.7	0.10	0.0206	0.144
$\mathrm{We}_0 = 100$	0.105	10.2	0.13	0.0189	0.138
$\mathrm{We}_0 = 200$	0.078	9.7	0.19	0.0166	0.129
$\mathrm{We}_0 = \infty$	0.034	8.0	0.40	0.0141	0.119
miscible	0.023	11.4	0.86	0.0198	0.141

5.3 Application to the simulation values

Table 1 applies (5.5) to the simulations of Refs. [24, 25, 26], averaged over $8 \le t/\tau_0 \le 9.4$, with θ_0 obtained from the mid-plane density fluctuation, $1 - \theta_0 = (2\rho_{\rm rms}/\Delta\rho)^2$.

Between $\mathrm{We}_0 = 50$ and $\mathrm{We}_0 = \infty$, the factor 3.84 in α splits as 1.76 from R_f^2, 1.49 from $1-\theta_0$ and 1.46 from $C_r^2\mathrm{Fr}_0^2$. Between $\mathrm{We}_0 = \infty$ and the miscible case, the factor 1.51 splits as $0.50 \times 4.25 \times 0.71$. The closure (5.6) would predict $4.25^{3/2} \approx 8.8$ for the latter, and a closure linear in $1-\theta_0$ with the other constants fixed would predict 4.3. The purity factor alone therefore overpredicts the difference between miscible and immiscible growth. The immiscible case without surface tension has a higher purity than the miscible case but the lowest R_f and $C_r\mathrm{Fr}_0$ of all five cases; surface tension raises R_f^2, the purity and $C_r^2\mathrm{Fr}_0^2$ by comparable factors. Within (5.5), the We_0 dependence is shared among the three factors, and R_f^2 is the largest.

5.4 Energy budget

Let $P_{\rm rel} = c_P\,\Delta\rho\,g h^2$ be the released potential energy per unit horizontal area ($c_P = 1/24$ for a linear profile), $K_E = c_K\bar{\rho}\,\dot{h}^2 h$ the kinetic energy per unit area, E_ε the dissipated energy and $S(t)$

the interface area per unit horizontal area. Conservation of energy, $P_{\rm rel} = K_E + E_\varepsilon + \sigma(S-1)$, together with $\dot{h}^2 = 4\alpha \mathrm{A_t} g h$, gives

$$\alpha = \frac{c_P}{2c_K}\,\frac{K_E}{P_{\rm rel}} = \frac{c_P}{2c_K}\big(1 - \Phi_\varepsilon - \Phi_\sigma\big)\,, \qquad \Phi_\varepsilon = \frac{E_\varepsilon}{P_{\rm rel}}\,, \quad \Phi_\sigma = \frac{\sigma(S-1)}{P_{\rm rel}} \geq 0\,. \tag{5.7}$$

The surface energy term lowers α. An increase of α with σ must therefore come from a smaller dissipated fraction Φ_ε or from a change in the shape ratio c_P/c_K. At $\mathrm{Re}_0 = \infty$, Φ_ε is the numerical dissipation of the scheme [6, 3].

6 Scaling of α with We_0

6.1 Complete self-similarity

Under the RC04 assumption that energy-containing quantities scale with h, the capillary group decays as $(\ell_\sigma/h)^2$ (Section 3.5), and (3.22) with $\nu = 2$ gives

$$\alpha(t) = \alpha_\infty\Big[1 - \frac{a_\sigma}{2\mathrm{A_t}\mathrm{We}_0}\Big(\frac{\lambda_0}{h}\Big)^2\Big] = \alpha_\infty\Big[1 - \frac{2\alpha\, a_\sigma}{\mathrm{We}(t)}\Big]\,. \tag{6.1}$$

The correction is proportional to We_0^{-1} (so linear in σ); it reduces α and decays as h^{-2}, and $\alpha_\infty = C_\mathrm{o}/4$ carries no We_0 dependence. The only other choice consistent with RC04 is a constant capillary group, $\Gamma_0 \propto \ell_\sigma^{-2}$ for the capillary moment $\langle c' f_z\rangle = (\sigma/\bar{\rho})\Gamma$. That shifts C_o by a pure number independent of λ_0 and again gives no We_0 scaling. Since $\mathrm{We}_0^{-1/2} = \ell_\sigma/\lambda_0$ at $\mathrm{A_t} = 0.5$, a $\mathrm{We}_0^{-1/2}$ scaling requires the capillary length to enter linearly ($\nu = 1$) at the energy-containing level. Self-similarity does not supply this; it requires a closure for C_o, which RC04 leave open.

6.2 A spectral closure beyond Ristorcelli & Clark

The closure in this subsection is not part of RC04. It uses the interface fractal dimension of Section 4.2.

Terms through which σ can act. By (3.5), $\langle c'c'\rangle_0 = 1$ for immiscible fluids, so σ can act on (3.13) only through the terms in Table 2. Here D_s is the fractal dimension of the interface, ℓ_in its inner cutoff, and the interface is assumed fractal between ℓ_H and h when estimating the surface energy. For $2 \leq D_s < 3$ the third term has the smallest exponent and is therefore the leading

Table 2: Terms through which surface tension can modify (3.13), with their magnitude relative to buoyancy.

term	relative magnitude	effect on α
capillary force in the flux budget, d_σ (3.15)	$(\ell_\sigma/h)^2$	decrease
surface energy in the energy budget, Φ_σ (5.7)	$(\ell_\sigma/h)^{(22-6D_s)/5}$	decrease
inner cutoff of the indicator spectrum	$(\ell_\mathrm{in}/h)^{3-D_s}$	increase

correction as $h/\ell_\sigma \to \infty$. It is also the only one that increases α.

Closure hypothesis. Only the indicator variance at the energy-containing scales drives the energy-containing flux. Flux produced at small scales is destroyed locally, as in the $k^{-7/3}$ range of the scalar-flux co-spectrum [15], and the mid-plane flux $F = 2\dot{h}/R_f$ is carried by the large scales. Accordingly, $\langle c'c'\rangle_0$ in (3.13) is replaced by $F_L\langle c'c'\rangle_0$, where F_L is the fraction of the variance in the energy-containing range.

Spectral fraction. For an interface of dimension D_s between ℓ_{in} and h, $E_c(k) \propto k^{D_s-4}$ by (4.4). Integrating over $1/h \le k \le 1/\ell_{\mathrm{in}}$,

$$F_L = \frac{\int_{1/h}^{2/h} E_c \,\mathrm{d}k}{\int_{1/h}^{1/\ell_{\mathrm{in}}} E_c \,\mathrm{d}k} = \frac{1 - 2^{D_s-3}}{1 - (\ell_{\mathrm{in}}/h)^{3-D_s}} \,, \tag{6.2}$$

and therefore

$$\alpha(h) = \frac{\alpha_\infty}{1 - (\ell_{\mathrm{in}}/h)^{3-D_s}} \approx \alpha_\infty\Big[1 + \Big(\frac{\ell_{\mathrm{in}}}{h}\Big)^{3-D_s}\Big] \,. \tag{6.3}$$

Without surface tension, an immiscible indicator has no inner cutoff, $\ell_{\mathrm{in}} \nrightarrow 0$; surface tension sets a finite ℓ_{in}. In the simulations, surface tension interrupts the inertial cascade: turbulent kinetic energy builds up near the Hinze scale, and the density spectra change accordingly [26].

Inner cutoff. Turbulence cannot deform the interface below ℓ_H (3.16), and buoyancy cannot deform it below ℓ_σ (2.3). The smaller of the two applies. By (3.17), $\ell_{\mathrm{in}} = \ell_\sigma$ for $h/\ell_\sigma \lesssim K_H^5$, a few hundred for $K_H = 3.3$, and $\ell_{\mathrm{in}} = \ell_H$ beyond. In the simulations, $K_H^5 \approx 60$ and h/ℓ_σ reaches 100–215, so ℓ_H is slightly smaller than ℓ_σ at late times; the two cutoffs differ by less than 25% (Figure 1), and the simulations cannot distinguish the two branches of (6.4), which for $D_s = 2$ give $\nu = 1$ and $\nu = 6/5$.

Result. Substituting (2.4) and (3.17) into (6.3),

$$\frac{\alpha - \alpha_\infty}{\alpha_\infty} \approx \begin{cases} (2\mathrm{A_t We_0})^{-(3-D_s)/2}\,\big(\lambda_0/h\big)^{3-D_s} \,, & h/\ell_\sigma \lesssim K_H^5 \,, \\ K_H^{3-D_s}\,(2\mathrm{A_t We_0})^{-3(3-D_s)/5}\,\big(\lambda_0/h\big)^{6(3-D_s)/5} \,, & h/\ell_\sigma \gtrsim K_H^5 \,. \end{cases} \tag{6.4}$$

For $D_s = 7/3$, this gives $\alpha - \alpha_\infty \propto \mathrm{We}_0^{-1/3}$ in the intermediate regime and $\propto \mathrm{We}_0^{-2/5}$ asymptotically. In the self-similar regime $h/\lambda_0 = \alpha_\infty(t/\tau_0)^2$, so at fixed t/τ_0 the same We_0 exponents hold, with time factors $(t/\tau_0)^{-4/3}$ and $(t/\tau_0)^{-8/5}$ respectively. A $\mathrm{We}_0^{-1/2}$ scaling requires $D_s = 2$ with $\ell_{\mathrm{in}} = \ell_\sigma$, or $D_s = 13/6$ with $\ell_{\mathrm{in}} = \ell_H$. In the form (4.1), (6.3) corresponds to $a < 0$ and $\nu = 3 - D_s$ or $\nu = \frac{6}{5}(3 - D_s)$. As $h/\ell_\sigma \nrightarrow \infty$, the correction vanishes, consistent with Section 3.6.

6.3 A correction linear in ℓ_σ/h

The case $\nu = 1$ has three equivalent forms. It is the capillary-velocity form (2.13), $\alpha - \alpha_\infty = K\ell_\sigma/h$; it is (4.1) with $\nu = 1$ and $a = -K/\alpha_\infty$; and it is the spectral closure (6.3) with $D_s = 2$ and $\ell_{\mathrm{in}} = \ell_\sigma$, for which $K = \alpha_\infty$. In terms of the instantaneous Weber number, by (2.12),

$$\alpha(t) - \alpha_\infty = K\Big(\frac{2\alpha}{\mathrm{We}(t)}\Big)^{1/2} = K\,(2\mathrm{A_t We_0})^{-1/2}\,\frac{\lambda_0}{h} \,. \tag{6.5}$$

At fixed h/λ_0, the correction is proportional to $\mathrm{We}_0^{-1/2}$, and it decays as h^{-1}. By Section 3.8, a correction of this form is not a transient of the RC04 balance with fixed constants, which relaxes as $h^{-R_f\hat{D}}$ with $R_f\hat{D} \approx 40$.

6.4 Comparison

Table 3 and Figure 3 summarize the exponents obtained above together with those inferred from the simulations in Section 7.

Table 3: Dependence of α on We_0 from Sections 2–6 and from the simulations (Section 7). Here ν is the exponent in $(\ell_\sigma/h)^\nu$.

approach	ν	We_0 dependence	effect of σ
complete self-similarity, (6.1)	2	$\alpha - \alpha_\infty \propto -\mathrm{We}_0^{-1}$	decrease
Hinze-scale estimate, (4.1)–(4.2)	$(7-3D_s)/5$	$\alpha - \alpha_\infty \propto -\mathrm{We}_0^{-\nu/2}$	decrease ($a>0$)
spectral closure, $\ell_{\mathrm{in}} = \ell_\sigma$, (6.4)	$3 - D_s$	$\alpha - \alpha_\infty \propto \mathrm{We}_0^{-(3-D_s)/2}$	increase
spectral closure, $\ell_{\mathrm{in}} = \ell_H$, (6.4)	$\frac{6}{5}(3-D_s)$	$\alpha - \alpha_\infty \propto \mathrm{We}_0^{-3(3-D_s)/5}$	increase
capillary velocity, (2.13)	1	$\alpha - \alpha_\infty \propto \mathrm{We}_0^{-1/2}$	increase ($K>0$)
simulations, fixed t/τ_0 (Table 4)	—	$\alpha \propto \mathrm{We}_0^{-0.37}$ to $\mathrm{We}_0^{-0.46}$	increase
simulations, fixed h/λ_0 (Figure 7)	0.97	$\alpha - \alpha_\infty \approx 8.9\,\ell_\sigma/h$	increase

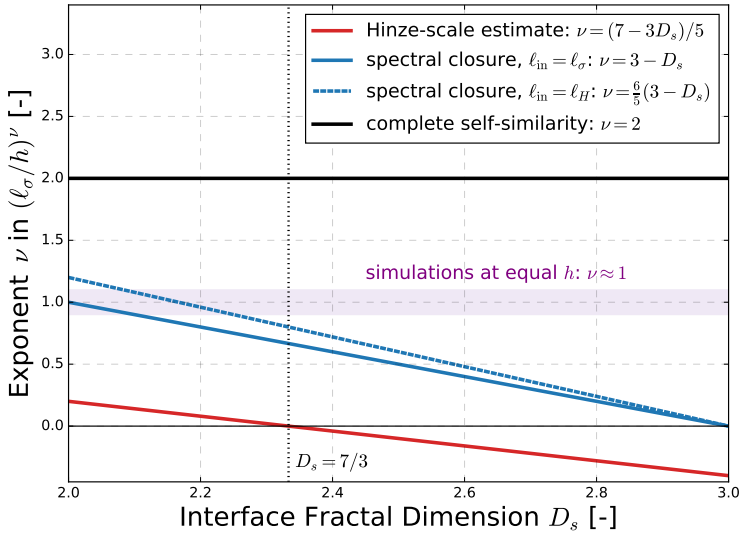
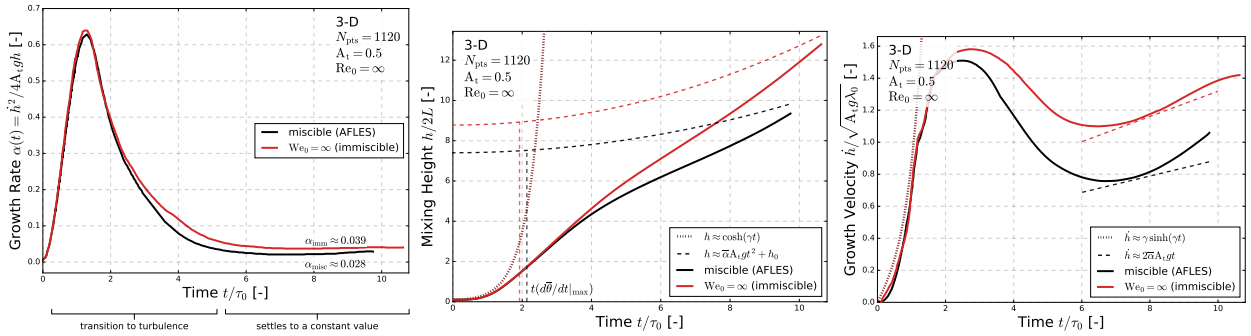


Figure 3: Exponent ν of the correction $(\ell_\sigma/h)^\nu$ as a function of the interface fractal dimension D_s: the Hinze-scale estimate (4.2), the spectral closure (6.4) with $\ell_{\mathrm{in}} = \ell_\sigma$ and $\ell_{\mathrm{in}} = \ell_H$, and complete self-similarity (3.18). The shaded band marks the value $\nu \approx 1$ obtained from the simulations at equal mixing width (Figure 7).

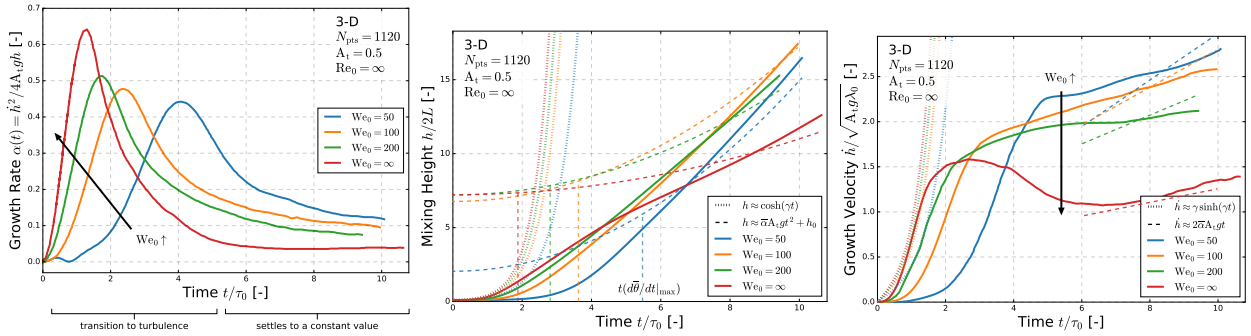
7 Simulation results

Figures 4 and 5 reproduce the growth parameter, mixing height and growth velocity presented in Refs. [23, 24]. Figures 6–8b are derived from the curves of Figure 5; the mixing height on its vertical axis satisfies $h/\lambda_0 = (\dot{h}/\sqrt{\mathrm{A_t} g \lambda_0})^2/4\alpha$ to within about 5%.



(a) (b) (c)

Figure 4: Miscible (AFLES) and immiscible ($We_0 = \infty$) simulations, 3-D, $N_{pts} = 1120$, $A_t = 0.5$, $Re_0 = \infty$ [23, 24]. (a) Growth parameter $\alpha(t)$ (2.6). (b) Mixing height with the linear-theory solution $h \propto \cosh(\gamma t)$ (dotted) and the fit $h \approx \bar{\alpha} A_t g t^2 + h_0$ (dashed); vertical lines mark the time of maximum $d\bar{\theta}/dt$. (c) Growth velocity $\dot{h}/(A_t g \lambda_0)^{1/2}$ with $\dot{h} \propto \gamma \sinh(\gamma t)$ (dotted) and $\dot{h} \approx 2\bar{\alpha} A_t g t$ (dashed).



(a) (b) (c)

Figure 5: Immiscible simulations at $We_0 = 50$, 100, 200 and ∞, 3-D, $N_{pts} = 1120$, $A_t = 0.5$, $Re_0 = \infty$ [23, 24]. Panels and line styles as in Figure 4.

7.1 Miscible and immiscible cases

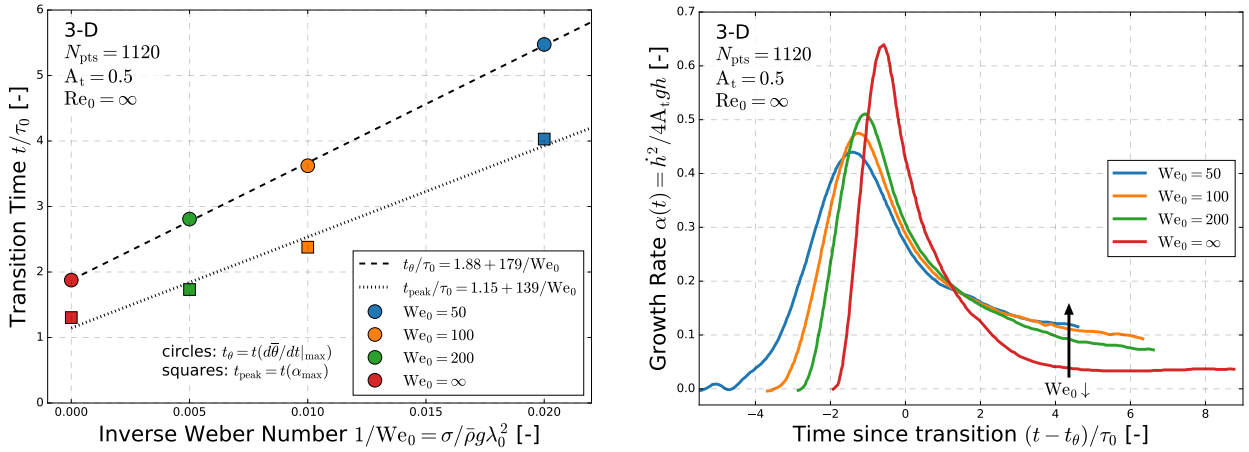
Figure 4 compares the miscible and immiscible cases without surface tension. The two follow the same linear and early nonlinear growth (Figures 4b and 4c) and the same peak in $\alpha(t)$ near $t \approx 1.3\tau_0$ (Figure 4a). After the transition, the immiscible layer grows faster, and $\alpha(t)$ settles at $\alpha_{imm} \approx 0.039$ compared with $\alpha_{misc} \approx 0.028$. In both cases, the late-time growth velocity increases approximately linearly in time (Figure 4c), as expected for quadratic growth.

7.2 Transition

Decreasing We_0 delays the transition and lowers the peak of $\alpha(t)$ (Figure 5a). The time of maximum $d\bar{\theta}/dt$ marked in Figure 5b, t_θ, and the time of peak α, t_{peak}, both increase linearly with We_0^{-1}, that is linearly with σ (Figure 6a):

$$\frac{t_\theta}{\tau_0} \approx 1.88 + \frac{179}{We_0}, \qquad \frac{t_{peak}}{\tau_0} \approx 1.15 + \frac{139}{We_0}, \tag{7.1}$$

with root-mean-square deviations of 0.03 and 0.13 over the four cases, including $\mathrm{We}_0 = \infty$. A delay linear in σ is the form expected from the first-order capillary correction to the growth rate (2.3). Aligned at t_θ (Figure 6b), the finite-We_0 curves nearly coincide for $(t - t_\theta)/\tau_0 \lesssim 3$; the ratio of the largest to the smallest α among $\mathrm{We}_0 = 50$–200 rises from 1.04 at $(t - t_\theta)/\tau_0 = 2$ to 1.32 at 4.5, and the ratio to $\mathrm{We}_0 = \infty$ from 1.3 to 2.8. Cook & Dimotakis [8] found that a progress variable brings cases with different initial conditions into closer agreement than time does, and RC04 attribute this to the virtual origin.



(a) (b)

Figure 6: (a) Transition time against $1/\mathrm{We}_0$: time of maximum $\mathrm{d}\bar{\theta}/\mathrm{d}t$ (circles) and of peak α (squares), with the fits (7.1). (b) Growth parameter against time since t_θ.

7.3 Comparison at fixed time

Table 4 lists the ratio $\alpha(\mathrm{We}_0{=}50)/\alpha(\mathrm{We}_0{=}200)$ at fixed t/τ_0 and the corresponding effective power law. At $t \approx 10\tau_0$, this is close to the fit $0.49\,\mathrm{We}_0^{-0.35}$ of Ref. [24]. The form $\alpha \approx \alpha_\infty + K\mathrm{We}_0^{-1/2}$ also fits at any single time, with $\alpha_\infty \approx 0.030$–$0.035$, but K drops from 1.2 at $6\tau_0$ to 0.62 at $9.4\tau_0$. The exponent obtained at fixed time is therefore not a property of the flow; it depends on when the simulations are compared.

Table 4: Growth parameter compared at fixed time, from the curves of Figure 5a.

t/τ_0	$\alpha(\mathrm{We}_0{=}50)/\alpha(\mathrm{We}_0{=}200)$	effective scaling
6.0	1.89	$\mathrm{We}_0^{-0.46}$
6.5	1.80	$\mathrm{We}_0^{-0.43}$
7.5	1.84	$\mathrm{We}_0^{-0.44}$
8.5	1.66	$\mathrm{We}_0^{-0.37}$
9.4	1.68	$\mathrm{We}_0^{-0.38}$

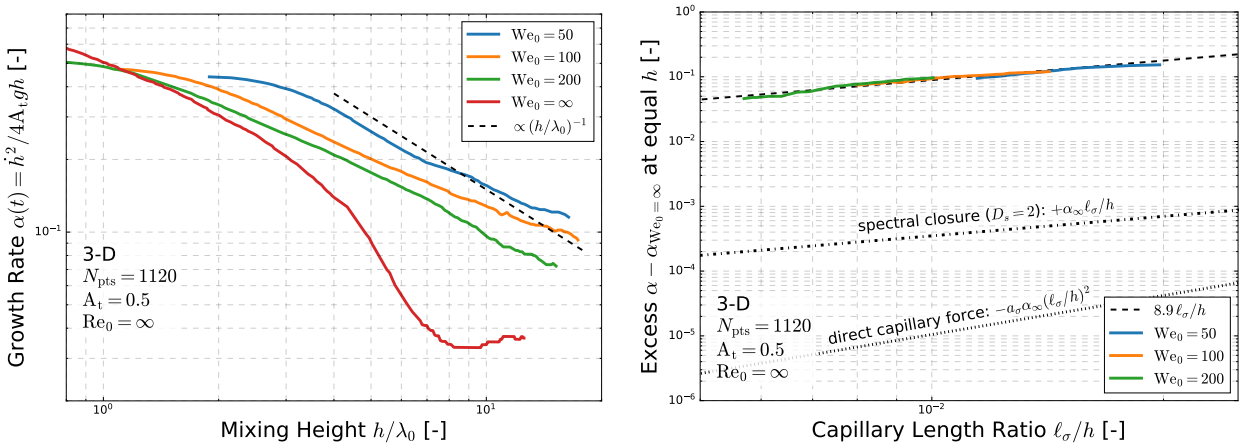
7.4 Comparison at equal mixing width

Figure 7a shows α against h/λ_0. Without surface tension, α settles at 0.034–0.036 for $h/\lambda_0 \gtrsim 8$. With surface tension, it continues to decrease, as $\alpha \propto (h/\lambda_0)^{-0.6}$ to $(h/\lambda_0)^{-0.85}$, and remains two

to five times larger at equal width. Over the range $7 \leq h/\lambda_0 \leq 12.5$, in which all four cases overlap, the excess $\alpha - \alpha_{\mathrm{We}_0=\infty}$ evaluated at equal h collapses onto a single line when plotted against ℓ_σ/h (Figure 7b):

$$\alpha - \alpha_{\mathrm{We}_0=\infty} \approx 8.9\,\frac{\ell_\sigma}{h}\,, \tag{7.2}$$

with the coefficient between 7.7 and 9.9 over all three Weber numbers and a fitted slope of 0.97 on logarithmic axes. This is the form (6.5) with $K \approx 8.9$: at fixed h/λ_0 the excess is proportional to $\mathrm{We}_0^{-1/2}$, and at fixed We_0 it decays as h^{-1}. Compared with the models of Section 6, the exponent agrees with the spectral closure for $D_s = 2$ and with the capillary-velocity form (2.13), while the magnitude is about 250 times that of the spectral closure ($K = \alpha_\infty$) and four orders of magnitude larger than the direct capillary correction (6.1), which has the opposite sign.



(a) (b)

Figure 7: (a) Growth parameter against mixing height after the peak of α. (b) Excess $\alpha - \alpha_{\mathrm{We}_0=\infty}$ at equal h against ℓ_σ/h for $7 \leq h/\lambda_0 \leq 12.5$, with the fit (7.2), the spectral closure (6.3) with $D_s = 2$, and the magnitude of the direct capillary correction (3.20) with $\nu = 2$ and $a_\sigma \approx 3$.

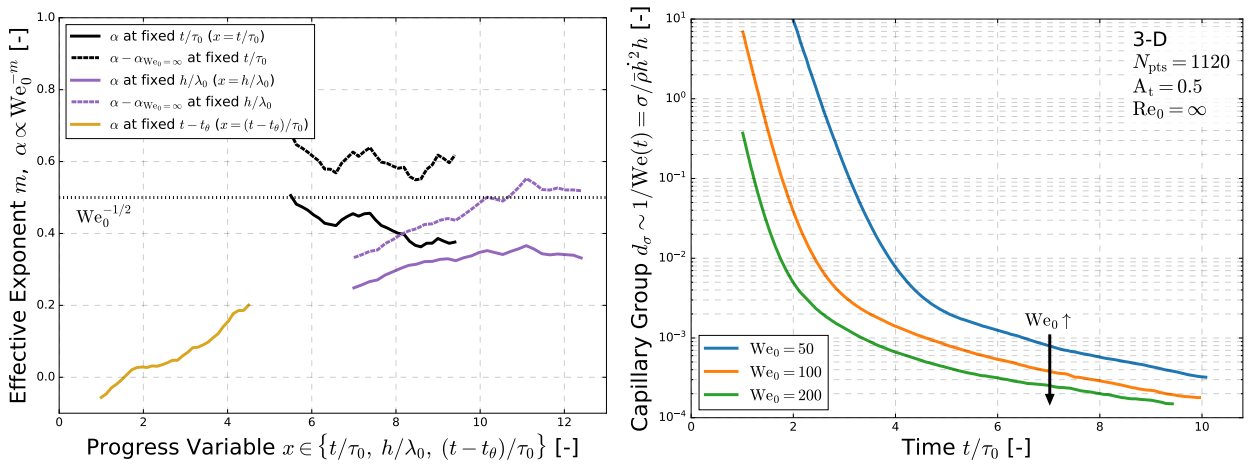
7.5 Dependence on the comparison

Figure 8a shows the effective exponent m in $\alpha \propto \mathrm{We}_0^{-m}$ across $\mathrm{We}_0 = 50$–200 for three comparisons. At fixed t/τ_0, m decreases from 0.51 to 0.38 between $5.5\tau_0$ and $9.4\tau_0$, and the excess over $\mathrm{We}_0 = \infty$ has $m \approx 0.55$–0.68. At fixed h/λ_0, m for α is 0.25–0.37, and for the excess it rises from 0.33 to about 0.5 as the $\mathrm{We}_0 = \infty$ case settles. At fixed time since transition, m rises from about zero to 0.2. The exponent reported for a given set of simulations therefore depends on the basis of comparison. The comparison at equal width is the natural one for the analyses of Sections 3–6, which are formulated in h.

Figure 8b shows the capillary group (3.15), $1/\mathrm{We}(t)$. It falls below 10^{-2} after the transition and is $(1.5\text{–}3.2) \times 10^{-4}$ at the end of the simulations, so the direct capillary term in the flux budget is negligible at late times, in agreement with Section 3.5.

7.6 Resolution of the capillary scale

The capillary length is close to the grid spacing in these simulations. Taking $\lambda_0 = L/k_{\max} \approx 1.02\,\mathrm{cm}$ and $\Delta x = L/N_{\mathrm{pts}} \approx 0.114\,\mathrm{cm}$, $\ell_\sigma/\Delta x = 1.26$, 0.89 and 0.63 for $\mathrm{We}_0 = 50$, 100 and 200, and the



(a) (b)

Figure 8: (a) Effective exponent m in $\alpha \propto \mathrm{We}_0^{-m}$, fitted over $\mathrm{We}_0 = 50$, 100, 200, for α and for $\alpha - \alpha_{\mathrm{We}_0=\infty}$ at fixed t/τ_0 and at fixed h/λ_0, and for α at fixed $t - t_\theta$; the horizontal axis is the corresponding progress variable, $x = t/\tau_0$, h/λ_0 or $(t - t_\theta)/\tau_0$. (b) Capillary group $d_\sigma \sim 1/\mathrm{We}(t)$ (3.15).

capillary cutoff wavelength ℓ_σ spans 4–8 $[\Delta x]$. That the excess is a single function of ℓ_σ/h (7.2) indicates that, over the simulated range, the late-time effect of surface tension is organized by ℓ_σ even though ℓ_σ is comparable to the grid spacing. The limit of drop breakup at the grid scale in these simulations is analyzed with a geometric model in Ref. [26].

8 Summary

1. Dimensional analysis gives $\alpha = \Psi(t/\tau_0; \mathrm{A_t}, \mathrm{We}_0, \mathrm{Re}_0, \dots)$ (2.9) and does not fix the We_0 dependence. A $\mathrm{We}_0^{-1/2}$ dependence is consistent with dimensional analysis as a correction $\alpha - \alpha_\infty = K\ell_\sigma/h$ (2.13), produced by a velocity scale $(\mathrm{A_t} g \ell_\sigma)^{1/2}$; it is not consistent with $\alpha \propto \mathrm{We}_0^{-1/2}$, which would vanish as $\sigma \to 0$. The instantaneous Weber number satisfies $\mathrm{We}(t) \propto \mathrm{We}_0 (t/\tau_0)^4$ and $(\ell_\sigma/h)^2 = 2\alpha/\mathrm{We}(t)$ (2.11)–(2.12).

2. For immiscible fluids, $\langle c'c' \rangle = 1 - C^2$ (3.4), so the mixedness vanishes and the RC04 constant becomes $C_\mathrm{o} = (1 - \chi_0)/\hat{D}$ (3.14): a buoyancy-efficiency factor divided by a normalized drag, both mid-plane statistics of the scalar-flux budget.

3. Under complete self-similarity, the capillary group is $1/\mathrm{We}(t) = (\ell_\sigma/h)^2/2\alpha$ (3.15), so α_∞ is independent of We_0. The first-order correction (3.19) has the form (4.1) with $\nu = 2$ and $a > 0$, and gives $\alpha - \alpha_\infty \propto -\mathrm{We}_0^{-1}$ (6.1). Departures from self-similar growth in the RC04 balance with fixed constants decay as $h^{-R_f \hat{D}}$, with $R_f \hat{D} \approx 40$ (3.24).

4. The modified growth equation (4.1) relates ν to the interface fractal dimension through (4.2), which follows from the indicator relations (4.4) and a Hinze-scale estimate of the capillary term (4.5); its effective growth parameter (4.3) is (3.21) with $\nu = 1/5$. With $a > 0$, both diagnostics decrease with σ, and $\nu > 0$ requires $D_s < 7/3$.

5. For the full width (2.5), the self-similar mass-flux ratio is $R_f = 8.0$–8.7 (5.4); the simulations

give 8.0 without surface tension and 9.7–10.7 with it. The decomposition (5.5) divides the increase of α with decreasing We_0 among comparable factors from R_f^2, the purity and $C_r^2\mathrm{Fr}_0^2$ (Table 1); the purity factor alone overpredicts the miscible–immiscible difference.

6. A spectral closure beyond RC04 gives an increase of α with σ, $\alpha - \alpha_\infty \propto \mathrm{We}_0^{-(3-D_s)/2}$ or $\mathrm{We}_0^{-3(3-D_s)/5}$ (6.4); $D_s = 2$ with $\ell_{\mathrm{in}} = \ell_\sigma$ gives $\nu = 1$ and $\mathrm{We}_0^{-1/2}$.

7. In the simulations, the transition time is linear in σ (7.1). At fixed time, the effective exponent drifts between 0.37 and 0.46 (Table 4). At equal mixing width, the excess over $\mathrm{We}_0 = \infty$ follows a single curve, $\alpha - \alpha_\infty \approx 8.9\,\ell_\sigma/h$ (7.2), which is a $\mathrm{We}_0^{-1/2}$ dependence at fixed h/λ_0 and corresponds to $\nu = 1$ and $a \approx -250$ in (4.1).

A Integration of the modified growth equation

Let $\xi = (\ell_\sigma/h)^\nu$. From (3.19), to first order in ξ,

$$\dot h = (\mathrm{A_t} g C_\mathrm{o} h)^{1/2}\Big[1 - \frac{a_\sigma}{2}\xi\Big] \,. \tag{A.1}$$

Then $t = \int \mathrm{d}h/\dot h$ gives, for $\nu \neq 1/2$,

$$t = 2\Big(\frac{h}{\mathrm{A_t} g C_\mathrm{o}}\Big)^{1/2}\Big[1 + \frac{a_\sigma}{2(1-2\nu)}\,\xi\Big] + t_v \,, \tag{A.2}$$

with t_v a constant of integration. Substituting (A.1) and (A.2) into $\alpha_{\mathrm{eff}} = \dot h/(2\mathrm{A_t} g t)$ and expanding,

$$\alpha_{\mathrm{eff}} = \frac{C_\mathrm{o}}{4}\Big[1 - a_\sigma\Big(\frac{1}{2} + \frac{1}{2(1-2\nu)}\Big)\xi - \frac{c_v}{\sqrt{h}}\Big] = \frac{C_\mathrm{o}}{4}\Big[1 - a_\sigma\,\frac{1-\nu}{1-2\nu}\,\xi - \frac{c_v}{\sqrt{h}}\Big] \,, \tag{A.3}$$

with $c_v = \frac{1}{2} t_v (\mathrm{A_t} g C_\mathrm{o})^{1/2}$, which is (3.21). Squaring (A.1) and dividing by $4\mathrm{A_t} g h$ gives (3.20).

List of symbols

Dimensions are given in terms of mass M, length L and time T; a dash marks a dimensionless quantity. The last column gives the equation, section, figure or table in which the symbol is defined or first used. A symbol with more than one meaning has one row per meaning.

Roman symbols

Symbol	Description	Dimensions	Defined in
a	Coefficient of the correction in the modified growth equation; $a < 0$ is required for α to increase with σ	–	(4.1)
a_σ	Coefficient of the first-order capillary correction, $a_\sigma = \hat d_\sigma/\hat D > 0$; the value of a in (4.1) for $\nu = 2$	–	(3.19)
$\mathrm{A_t}$	Atwood number, $\Delta\rho/(2\bar\rho) = (\rho_h - \rho_\ell)/(\rho_h + \rho_\ell)$; 0.5 in the simulations	–	(2.1)
$\mathrm{A_{eff}}$	Effective Atwood number of CCM04, $\mathrm{A_t}(1-\theta_0)^{1/2}$	–	§5.2

Symbol	Description	Dimensions	Defined in
c	Phase indicator, $+1$ in the heavy fluid and -1 in the light fluid, so that $\rho = \bar{\rho}(1 + \mathrm{A_t} c)$	–	§3.1
c'	Fluctuation of the indicator about its plane average, $c' = c - C$	–	§3.1
$\hat{c}_{\mathbf{k}}$	Fourier coefficient of c' at wavevector $\mathbf{k}$	–	§3.4
c_K	Shape coefficient of the kinetic energy, $K_E = c_K \bar{\rho}\, \dot{h}^2 h$	–	(5.7)
c_P	Shape coefficient of the released potential energy, $P_{\mathrm{rel}} = c_P\, \Delta\rho\, g h^2$; $1/24$ for a linear $\bar{V}$ profile	–	(5.7)
c_v	Virtual-origin coefficient in α_{eff}, $c_v = \frac{1}{2} t_v (\mathrm{A_t} g C_{\mathrm{o}})^{1/2}$	$\mathrm{L}^{1/2}$	(3.21), (A.3)
C	Plane-averaged indicator, $C = \langle c \rangle = 2\bar{V} - 1$; $C = 0$ on the mid-plane	–	§3.1
C_{o}	Growth constant of RC04, $\dot{h}^2 = \mathrm{A_t} g C_{\mathrm{o}} h$; the constant amplitude of the mean indicator in the self-similar form; $C_{\mathrm{o}} = 4\alpha_\infty$	–	(3.14)
C_H	Constant in the Hinze scale, $C_H \approx 0.725$	–	(3.16)
C_r	Correlation coefficient of the mid-plane mass flux, $\langle \rho' w' \rangle_0 = -C_r \rho_{\mathrm{rms}} w_{\mathrm{rms}}$ (CCM04 eq. 7.5)	–	§5.2
C_w	Constant in the CCM04 velocity closure, $w_{\mathrm{rms}} = C_w (\mathrm{A_{eff}} g h)^{1/2}$	–	(5.6)
C_ε	Dissipation coefficient, $\varepsilon_0 = C_\varepsilon \dot{h}^3 / h$	–	§3.5
d_μ	Normalized mid-plane sink of F from viscosity, $(h/\dot{h}^2)(\mu/\bar{\rho}) \langle c' \nabla^2 w \rangle_0$	–	(3.11)
d_N	Normalized mid-plane sink of F from the nonlinear pressure p_N	–	(3.11)
d_T	Normalized mid-plane sink of F from turbulent transport, $\partial \langle w w c' \rangle / \partial z$	–	(3.11)
d_σ	Capillary group: normalized mid-plane sink of F from the solenoidal capillary force; $d_\sigma \sim 1/\mathrm{We}(t)$	–	(3.11), (3.15)
$\hat{d}_\sigma$	Amplitude of the capillary group, $d_\sigma = \hat{d}_\sigma (\ell_\sigma / h)^\nu$	–	(3.18)
$\hat{D}$	Normalized inertia and drag in the mid-plane flux budget, $\hat{D} = 1/R_f + d_T + d_N + d_\mu - \hat{w}\, f_C'(0)$	–	(3.12)
D_s	Fractal dimension of the interface; 2 for a smooth interface and 7/3 for an indicator with the Obukhov–Corrsin spectrum	–	(4.4)
$\mathbf{e}_z$	Upward unit vector	–	(3.1)
E	Constant of integration in the relaxation solution	$\mathrm{L}^{2+\beta}\mathrm{T}^{-2}$	(3.24)
$E_c(k)$	Wavenumber spectrum of the indicator, $E_c \propto k^{-p}$	L	§4.2
E_ε	Kinetic energy dissipated per unit horizontal area	$\mathrm{M\,T}^{-2}$	(5.7)
$\mathbf{f}$	Capillary force per unit mass, $\mathbf{f} = \nabla \cdot \mathbf{T}^\sigma = (\sigma/\bar{\rho})\, \kappa\, \hat{\mathbf{n}}\, \delta_s$; f_z is its vertical component	$\mathrm{L\,T}^{-2}$	§3.1
$f(\eta)$	Similarity profile of a self-similar quantity, $F(z,t) = F_{\mathrm{o}}(t)\, f(\eta)$	–	§3.2
$f_C(\eta)$	Similarity profile of the mean indicator, $C = f_C(\eta)$; $f_C'(0) = h\, \partial C / \partial z \vert_0$	–	§3.3

Symbol	Description	Dimensions	Defined in
F	(i) Mid-plane indicator flux, $F = -\langle wc' \rangle_0 = 2\dot{h}/R_f$	$\mathrm{L\,T^{-1}}$	§3.4
	(ii) Generic self-similar quantity, $F(z,t) = F_\mathrm{o}(t)\, f(\eta)$, with temporal amplitude $F_\mathrm{o}(t)$	varies	§3.2
F_L	Fraction of the indicator variance in the energy-containing range $1/h \le k \le 2/h$	–	(6.2)
Fr_0	Mid-plane Froude number, $\mathrm{Fr}_0^2 = w_\mathrm{rms}^2/(\mathrm{A_t} g h)$	–	§5.2
g	Gravitational acceleration, acting in the $-z$ direction	$\mathrm{L\,T^{-2}}$	§2.1
h	Full mixing width, $h = 6\int \bar{V}(1-\bar{V})\,\mathrm{d}z$; the total width of a linear profile	L	(2.5)
$\dot{h}$, $\ddot{h}$	Growth velocity and acceleration of the mixing width	$\mathrm{L\,T^{-1}}$, $\mathrm{L\,T^{-2}}$	(2.6)
h_0	Width at $t = 0$ in the RC04 solution; offset of the quadratic fit in Figures 4 and 5	L	(2.7)
h_C, $\dot{h}_C$	Entrainment length of CCM04, $h_C = 4\int_{-\infty}^{0} \bar{V}\,\mathrm{d}z$, and its growth velocity	L, $\mathrm{L\,T^{-1}}$	§5.1, (5.4)
$\mathbf{I}$	Identity tensor	–	(3.3)
k, $\mathbf{k}$	Wavenumber and wavevector; in (2.3) $k = 2\pi/\text{wavelength}$; k_z is the vertical component of $\mathbf{k}$	$\mathrm{L^{-1}}$	(2.3)
k_min, k_max	Smallest and largest mode numbers of the initial perturbation in a domain of width L; $\lambda_0 = L/k_\mathrm{max}$	–	(2.2)
K	(i) Coefficient of the correction linear in ℓ_σ/h, $\alpha - \alpha_\infty = K\ell_\sigma/h$; $K \approx 8.9$ in the simulations	–	(2.13), (7.2)
	(ii) Coefficient of the fixed-time fit $\alpha \approx \alpha_\infty + K\mathrm{We}_0^{-1/2}$	–	§7.3
K_E	Kinetic energy per unit horizontal area, $K_E = c_K \bar{\rho}\,\dot{h}^2 h$	$\mathrm{M\,T^{-2}}$	(5.7)
K_H	Prefactor of the Hinze scale, $K_H = C_H\, C_\varepsilon^{-2/5} (2\alpha)^{-3/5} \approx 3.3$	–	(3.17)
ℓ_H	Hinze scale: the smallest interface scale that turbulence can deform	L	(3.16)
ℓ_in	Inner cutoff of the fractal range of the interface; the smaller of ℓ_σ and ℓ_H	L	§6.2
ℓ_σ	Capillary length, $(\sigma/\Delta\rho\, g)^{1/2}$; the inverse of the cutoff wavenumber of linear theory	L	(2.3)
L	Width of the computational domain	L	(2.2)
m	Effective exponent in $\alpha \propto \mathrm{We}_0^{-m}$	–	Fig. 8
$\hat{\mathbf{n}}$	Unit normal to the interface, $\nabla c/\lvert\nabla c\rvert$	–	(3.3)
N_pts	Number of grid points in each direction, 1120	–	Fig. 4
p	(i) Pressure divided by $\bar{\rho}$; p' is its fluctuation, split into buoyancy, nonlinear and capillary parts p_B, p_N and p_σ	$\mathrm{L^2T^{-2}}$	(3.1), (3.9)
	(ii) Spectral exponent of the indicator, $E_c \propto k^{-p}$; $p = 2q + 1 = 4 - D_s$	–	(4.4)

Symbol	Description	Dimensions	Defined in
P_{rel}	Potential energy released per unit horizontal area	$\mathrm{M\,T^{-2}}$	(5.7)
q	Exponent of the second-order structure function, $\mathcal{S}_2 \propto r^{2q}$; $q = (3 - D_s)/2$	–	(4.4)
r, $\mathbf{r}$	Separation distance and separation vector	L	§4.2
R_f	Mass-flux shape factor, $R_f = \Delta\rho\,\dot{h}/\lvert\langle\rho' w'\rangle_0\rvert$; $2/R_f = \int_{-\infty}^{0}(1 + f_C)\,\mathrm{d}\eta$, so that $\langle wc'\rangle_0 = -2\dot{h}/R_f$; 8.0–8.7 for self-similar profiles; twice the scaling ratio of Refs. [24, 25]	–	(5.3), (3.7)
Re_0	Reynolds number of the initial perturbation; $\mathrm{Re}_0 = \infty$ (no physical viscosity) in the simulations	–	(2.9)
$S(t)$	Interface area per unit horizontal area	–	(5.7)
t	Time	T	–
t_θ	Transition time: the time of maximum $\mathrm{d}\bar{\theta}/\mathrm{d}t$	T	(7.1)
t_{peak}	Time of the peak of $\alpha(t)$	T	(7.1)
t_v	Constant of integration (virtual origin) in $t(h)$	T	(A.2)
t_σ	Capillary time, $(\bar{\rho}\lambda_0^3/\sigma)^{1/2}$	T	(2.10)
$\mathbf{T}^\sigma$	Capillary stress tensor divided by $\bar{\rho}$	$\mathrm{L^2T^{-2}}$	(3.3)
$\mathbf{u}$	Velocity, $\mathbf{u} = (u, v, w) = (u_1, u_2, u_3)$; u_i in index notation; $u_3 = w$ in the notation of RC04	$\mathrm{L\,T^{-1}}$	(3.1)
U_σ	Velocity scale formed from capillarity and buoyancy, $(\mathrm{A_t} g \ell_\sigma)^{1/2}$	$\mathrm{L\,T^{-1}}$	(2.13)
V	Volume fraction of the heavy fluid; $\bar{V}(z,t)$ is its plane average and $\bar{V}_0$ the mid-plane value	–	(2.5)
w	Vertical (upward) velocity component; w_{rms} is its root-mean-square value on the mid-plane	$\mathrm{L\,T^{-1}}$	§3.1
$\hat{w}$	Normalized mid-plane variance of the vertical velocity, $\langle ww\rangle_0/\dot{h}^2$	–	(3.11)
We_0	Initial Weber number, $\bar{\rho}\, g\, \lambda_0^2/\sigma$	–	(2.2)
$\mathrm{We}(t)$	Instantaneous Weber number, $\bar{\rho}\,\dot{h}^2 h/\sigma$	–	(2.11)
x	Progress variable in Figure 8: t/τ_0, h/λ_0 or $(t - t_\theta)/\tau_0$	–	Fig. 8
$\mathbf{x}$, x_i	Position vector and its components	L	(3.9)
Δx	Grid spacing, L/N_{pts}	L	§7.6
y	Square of the growth velocity, $y = \dot{h}^2$	$\mathrm{L^2T^{-2}}$	(3.23)
Y	Mass fraction of the heavy fluid	–	§2.2
z	Vertical coordinate, positive upward; $z = 0$ is the mid-plane	L	§2.1

Greek symbols

Symbol	Description	Dimensions	Defined in
α	Growth parameter in $h \approx \alpha \mathrm{A_t} g t^2$	–	§1
$\alpha(t)$	Instantaneous growth parameter, $\dot{h}^2/(4\mathrm{A_t} g h)$; the measure used in the simulations	–	(2.6)

Symbol	Description	Dimensions	Defined in
α_{eff}	Effective growth parameter of RC04, $\mathrm{d}h/\mathrm{d}(\mathrm{A_t}gt^2) = \dot{h}/(2\mathrm{A_t}gt)$	–	(2.6)
α_∞	Self-similar limit of α as $h/\ell_\sigma \to \infty$, $\alpha_\infty = C_\mathrm{o}/4$	–	(3.14)
α_{imm}, α_{misc}	Late-time α of the immiscible ($\mathrm{We}_0 = \infty$) and miscible simulations, ≈ 0.039 and 0.028	–	§1
$\alpha_{\mathrm{We}_0=\infty}$	$\alpha(t)$ of the immiscible simulation without surface tension, taken at the same h	–	(7.2)
$\bar{\alpha}$	Growth parameter of the quadratic fit $h \approx \bar{\alpha}\mathrm{A_t}gt^2 + h_0$	–	Fig. 4
β	Exponent of the decay of departures from $\dot{h}^2 \propto h$, $\beta = R_f\hat{D} - 1$	–	(3.23)
γ	Linear growth rate of a mode of wavenumber k	T^{-1}	(2.3)
Γ	Capillary moment per unit $\sigma/\bar{\rho}$, $\langle c' f_z \rangle = (\sigma/\bar{\rho})\Gamma$; Γ_0 is its mid-plane value	L^{-2}	§6.1
δ_s	Surface delta function, concentrated on the interface	L^{-1}	(3.3)
$\Delta\rho$	Density difference, $\rho_h - \rho_\ell$	$\mathrm{M\,L}^{-3}$	(2.1)
ε	Dissipation rate of turbulent kinetic energy per unit mass; ε_0 is its mid-plane value	$\mathrm{L}^2\mathrm{T}^{-3}$	(3.16)
η	Similarity variable, $\eta = z/h$	–	§3.3
θ	Mixedness; θ_0 is its mid-plane value	–	§3.2
$\bar{\theta}$	Global mixedness of the layer, whose maximum rate of change defines t_θ	–	Fig. 4
θ_m	Molecular mixing parameter of RC04, $1 - \langle c'c' \rangle/[(1-C)(1+C)]$; zero for immiscible fluids	–	§3.2
κ	Curvature of the interface, $\kappa = -\nabla \cdot \hat{\mathbf{n}}$	L^{-1}	§3.1
λ_0	Initial perturbation length: the wavelength of mode $k_{\max}$, $\lambda_0 = L/k_{\max}$	L	(2.2)
μ	Dynamic viscosity; $\mu/\bar{\rho}$ is the kinematic viscosity	$\mathrm{M\,L}^{-1}\mathrm{T}^{-1}$	(3.1)
ν	Exponent of the correction $(\ell_\sigma/h)^\nu$; 2 under complete self-similarity, $(7-3D_s)/5$ from the Hinze-scale estimate and ≈ 1 in the simulations	–	(3.18), (4.1)
ξ	Expansion variable, $\xi = (\ell_\sigma/h)^\nu$	–	App. A
ρ	Density; $\langle\rho\rangle$ is its plane average and ρ' its fluctuation	$\mathrm{M\,L}^{-3}$	§3.1
ρ_h, ρ_ℓ	Densities of the heavy and light fluids	$\mathrm{M\,L}^{-3}$	§2.1
$\bar{\rho}$	Mean density, $\frac{1}{2}(\rho_h + \rho_\ell)$	$\mathrm{M\,L}^{-3}$	(2.1)
ρ_{rms}	Root-mean-square density fluctuation on the mid-plane, $\Delta\rho\,[\bar{V}_0(1-\bar{V}_0)(1-\theta_0)]^{1/2}$	$\mathrm{M\,L}^{-3}$	§5.2
σ	Surface tension coefficient	$\mathrm{M\,T}^{-2}$	§2.1
Σ	Surface density: interface area per unit volume; $\Sigma(r)$ is its value measured at resolution r	L^{-1}	§3.5
τ_0	Initial time scale, $(\lambda_0/\mathrm{A_t}g)^{1/2}$	T	(2.2)
Φ	Dimensionless function giving h/λ_0	–	(2.9)
Φ_ε	Fraction of the released potential energy that is dissipated, $E_\varepsilon/P_{\mathrm{rel}}$	–	(5.7)

Symbol	Description	Dimensions	Defined in
Φ_σ	Fraction of the released potential energy stored as surface energy, $\sigma(S-1)/P_{\mathrm{rel}}$	–	(5.7)
χ	Fraction of the buoyancy production cancelled by the buoyancy pressure; χ_0 is its mid-plane value; 0 for vertical columns, 1/3 for isotropic structures, 1 for horizontal layers	–	(3.10)
Ψ	Dimensionless function giving $\alpha(t)$	–	(2.9)

Operators and calligraphic symbols

Symbol	Description	Defined in
$\langle\cdot\rangle$	Average over a horizontal plane	§3.1
$(\cdot)'$	Fluctuation about the plane average	§3.1
$\dot{(\cdot)}$, $\ddot{(\cdot)}$	First and second time derivatives	(2.6)
$(\cdot)\vert_0$	Evaluated on the mid-plane $z=0$	(3.11)
∇^{-2}	Inverse Laplacian	(3.10)
$\mathcal{O}(\cdot)$	Order of magnitude	§3.5
$\mathcal{P}$	Projection onto the solenoidal (divergence-free) part, $\mathcal{P}\mathbf{f} = \mathbf{f} - \nabla p_\sigma$	§3.4
$\mathcal{S}_2(r)$	Second-order structure function of the indicator, $\langle[c(\mathbf{x}+\mathbf{r}) - c(\mathbf{x})]^2\rangle$	§4.2

Subscripts and accents

Symbol	Description	Defined in
$(\cdot)_{\mathrm{o}}$	Temporal amplitude in the RC04 self-similar form $F(z,t) = F_{\mathrm{o}}(t)\, f(\eta)$	§3.2
$(\cdot)_0$	On a statistic: value on the mid-plane $z=0$. On λ_0, τ_0, We_0, Re_0 and h_0: initial or reference value	§3.2
$(\cdot)_h$, $(\cdot)_\ell$	Heavy and light fluid	§2.1
$(\cdot)_\infty$	Self-similar limit $h/\ell_\sigma \to \infty$; on We_0, the case without surface tension	(3.14)
$(\cdot)_B$, $(\cdot)_N$, $(\cdot)_\sigma$	Buoyancy, nonlinear and capillary parts of the pressure	§3.4
$(\cdot)_T$, $(\cdot)_\mu$	Turbulent-transport and viscous terms of the flux budget	(3.11)
$(\cdot)_C$	On f_C: mean indicator. On h_C: entrainment length of CCM04	§3.3
$(\cdot)_H$	Hinze scale	(3.16)
$(\cdot)_L$	Large (energy-containing) scales	(6.2)
$(\cdot)_{\mathrm{in}}$	Inner cutoff	§6.2
$(\cdot)_{\mathrm{eff}}$	Effective	(2.6)
$(\cdot)_{\mathrm{imm}}$, $(\cdot)_{\mathrm{misc}}$	Immiscible and miscible simulations	§1

Symbol	Description	Defined in
$(\cdot)_{\text{peak}}$	At the peak of $\alpha(t)$	(7.1)
$(\cdot)_{\text{rel}}$	Released	(5.7)
$(\cdot)_{\text{rms}}$	Root mean square	§5.2
$(\cdot)_s$	Surface (δ_s, D_s)	(3.3)
$(\cdot)_v$	Virtual origin	(A.2)
$(\cdot)_z$	Vertical component	§3.1
$(\cdot)_\theta$	Defined by the mixedness	(7.1)
$\overline{(\cdot)}$	On V: plane average. On θ: global value. On ρ: mean of the two densities. On α: fitted value	(2.5)
$\hat{(\cdot)}$	Normalized quantity ($\hat{D}$, $\hat{d}_\sigma$, $\hat{w}$), unit vector ($\hat{\mathbf{n}}$) or Fourier coefficient ($\hat{c}_{\mathbf{k}}$)	(3.11)

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